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Chapter One
Charge, Electric field & Gauss's Law

1. What is Electrostatic induction?
2. Define: Charge, electricity.
3. What is meant by Quantization of charge?
4. What is electric field?
5. What is meant by electric field intensity?
6. Write the expression for electric field intensity due to point charge?
7. What is electric dipole?
8. What is meant by electric dipole moment?
9. Calculate electric field intensity (E) for an electric dipole.
10. What is meant by charge density?
11. Write the expression of electric field intensity due to a ring of charge q and of radius a.
12. Deduce an expression of electric field intensity due to a long straight, uniformly charged wire. Or, Figure shows a section of an infinite line of a charge whose linear charge density (that is, the charge per unit length, measured in coul /meter) has the constant value λ. Calculate the field E a distance y from the line.
13. Expression for Torque and potential energy of an electric dipole in a uniform electric field. Or, Derive the Equation of Torque and stored energy when an electric dipole is placed in a uniform electric field.
14. What is line of forces?
15. Properties of electric lines of force.
16. Define: Electric Flux, Electric Potential.
17. State and explain Coulomb's law.
18. State and explain Gauss's law.
19. What is meant by Gaussian surface?
20. Deduce Coulomb's law from Gauss's law.
21. Write down the Application of Gauss's law.
22. An infinite length rod have linear charge density λ. Calculate E for the rod at a distance 'r' from the rod. Or, Find an expression of $\vec{E}$ for a rod (sylindrical surface).
23. Derive an expression for Potential due to a point charge. Or, Potential due to a point charge.
24. What is electric potential energy?
25. What do you understand by equipotential surface?
26. What is meant by potential difference?
27. Derive an expression for potential due to point charge?
28. Derive an expression for electric potential due to a charged disk.
29. Derive an expression for potential due to a dipole.
30. What do you understand by Capacitance?
31. What is Capacitor?
32. What is Parallel plate Capacitor?
33. Calculate Equivalent Capacitance for Series and parallel combination.

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34. What is dielectric?
35. What do you understand by dielectric constant (k)?
36. What do you mean by electric Vector?
37. Deduce the expression for energy stored in an electric field.
38. Deduce the expression for energy stored of a charged capacitor.
39. Calculate Capacitance for parallel plate capacitor (with and without Dielectric constant).

Chapter Two
Electric current, simple circuits and electric measurement

1. What is electric current?
2. What do you understand by current? Find an expression for current?
3. What do you understand by current density? Find an expression for current density?
4. What is Resistance?
5. What is conductance?
6. State and explain ohm's law?
7. Find out the expression energy/power (p) in an electric circuit.
8. What do you understand by resistivity?
9. State and explain Kirchhoff's law.
10. Calculate Equivalent resistance for Series and parallel connection circuit.
11. Define: Electromotive force.
12. For an RC circuit, when the switch is connected to the battery. Find the expression for charge q and current I. Or, Prove $q = q_0(1 - e^{-\frac{t}{Rc}})$ Or, $q = CV(1 - e^{-\frac{t}{Rc}})$
13. Establish Wheatstone Bridge Principle to measure resistance. OR, what are Kirchhoff's? apply these law's to find a relation between the resistances of the four arm's in Wheatstone bridge when there is no current in flowing through the galvanometer.
14. Write a short note with figure and application for the following instruments:
    - i. Potentiometer
    - ii. Moving coil galvanometer
    - iii. Ammeter
    - iv. Voltmeter
    - v. Multimeter
    - vi. Wattmeter

Chapter Three
Magnetic field & force on current

1. What do you understand by magnetic dipole?
2. Define: Magnetic Field, Magnetic field strength, magnetic flux.
3. Difference between Electric and Magnetic field.

4. Show that magnetic force acting on a current carrying conductor of length L in $\vec{F} = I(\vec{L} \times \vec{B})$, where symbols have their usual meaning. Or, deduce the expression of force on a current carrying conductor.
5. Find out the expression of torque and potential energy of a current loop placed in a magnetic field.
6. What is Lorentz force?
7. State and explain Fleming’s left hand rule and Fleming's right-hand rule.
8. Difference between Fleming’s Left-Hand and Right-Hand Rule.
9. State and explain Hall Effect and Hal voltage.
10. State and explain Amperes law.
11. Calculate the magnetic field $'B'$ for a solenoid or a pack coil or a packed current loop by using Amperes law.
12. Explain Biot- Savart Law and express it in vector form.
13. Find the expression B by using Biot Savart law for a long straight wire.
14. Find the expression B by using Biot Savart law for a circular wire.

Chapter Four
Magnetic properties of matter

1. State and explain Faraday’s law of induction.
2. What is lenz’s law?
3. What is inductance?
4. Define self-inductance and mutual-inductance
5. Distinguish between self-inductance and mutual-inductance.
6. Calculate the value of self-inductance for toroid.
7. Calculate the value of self-inductance for solenoid.
8. Derive the transient response of an RL circuit for charging and discharging phase.
9. Prove that , Energy stored in magnetic field is $U = \frac{1}{2} Li^2$
10. What is a magnetic dipole?
11. What is pole?
12. Define magnetization?
13. State and explain Gauss’s law for magnetism.
14. State and explain coulomb’s law for magnetism / magnetic induction.
15. Write down (at least 6) characteristics of Paramagnetic, Diamagnetic and ferromagnetic material.
16. Write down the difference between Paramagnetic, Diamagnetic and ferromagnetic material.
17. What do you mean by Hysteresis and eddy current losses?
18. Describe hysteresis curve of magnetization.
19. What is the difference between line of force and line of induction?